Exam Date & Time: 07-Jan-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

3rd SEM B. Tech (BME) DEGREE END SEMESTER MAKE UP EXAMINATIONS, JAN-2025

DIGITAL SYSTEM DESIGN [BME 2123]

Marks: 50 **Duration: 180 mins.**

Des

Answer all the questions. Section Duration: 180 mins

1A) Explain Serial Input Serial Output (SISO) with the help of D-flip flops and provide the corresponding Truth table and timing waveform representation. (4)

1B) Illustrate Master-Slave J-K Flip flop using NAND gates and provide the corresponding timing waveform. (4)

1C) Explain the functionality of a 1-bit comparator using a logic diagram and provide the associated truth table. (2)

2A) Illustrate and analyze the structure of a standard cell, detailing its components and necessary features with a clear diagram. (4)

2B) Examine the structure of a gate array with a detailed diagram, explaining its essential components and features. (4)

2C) Compare and differentiate two key aspects between CPLD and FPGA architectures. (2)

3A) Analyze and implement the following function using a Programmable Logic Array (PLA)

$$A = XY + X\overline{Z}$$

$$B = X\overline{Y} + YZ + X\overline{Z}$$

(5)

3B) Analyze and implement the following function using Programmable Array Logic (PAL)

$$F1 = \sum m(3,5,7) \text{ and } F2 = \sum m(4,5,7)$$

(3)

3C) Analyze the purpose and applications of custom chips in electronic design. (2)

4A) Analyze and illustrate the implementation of F(A,B,C) =∑m(0,2,3,7) and G(A,B,C) = ∑m(1,4,6,7) using Active high and Active low decoder (4)

4B)	Design a two-input CMOS NOR gate with the help of a neat circuit and the corresponding truth table considering the details of pull-up and pull-down transistors	(4)
4C)	Analyze and draw the minimum CMOS transistor network that realizes the functionality of the given Boolean equation $F = \overline{((A+B)\,C + D)}$	(2)
5A)	Explain the process, with detailed steps, of converting a JK flip-flop into a D flip-flop. Include the corresponding Boolean expressions and a circuit diagram	(4)
5B)	Analyze and illustrate the general structure of a Field Programmable Gate Array (FPGA) with a suitable diagram.	(4)
5C)	Design a two-input NMOS AND gate with the help of a neat circuit and corresponding truth table	(2)

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